

Bitwise Operators

Bitwise Operators: ye hume bit manipulation krne me help krte ha.

Bit manipulation: it means ki ye (0) or (1) ke level pr jaa ke kaam krte ha.

LIST OF BITWISE OPERATORS

1. '&' Bitwise and
2. '|' Bitwise or
3. '^' Bitwise exclusive or
4. '~' Bitwise not

Lets take an example: byte a = 2;

byte b = 3;

%, bit value of, a = 00000010

B = 00000011

Truth Table					
A	B	A & B (both 1 chahiye)	A B (ek 1 bhi chlega)	A ^ B (odd no of 1 chiye)	~A (reverse of a)
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

5. '<<' shift left: Isme no left shift hoga or values double hone lagti ha.

6.

Example: byte b = 7; (to yaha pr sabse phle byte hamara type promote hoga int me.)

⇒ b = << 1 (to hamari 7 ki sabhi binary values 1-1 step left side shift ho jaygi.)

Or agar ab hm krte ha b = << 34 to vo mode ho jayga 32 se . or value aaygi 2 to hamri values 2 steps shift ho jaygi.

7. '>>' shift right
8. '>>>' shift right with 0 fill

Logical Operators

Ye boolean ke sath use hote ha jase ki Relational operators hote the.

1. **&&** (and logical operator)
2. **||** (or logical operator)
3. **!** (not logical operator)

Yha pr 1 or concept ata ha **SHORT-CIRCUIT**

FOR example: `int a = 5;`

```
int b = 10;
```

```
int c = 15;
```

```
boolean d = (a < b) && (a < c);
```

Yha pr hume pta ha ki && ka lia hume dono conditions true chiye or agar kisi tarah se hme phle hii pta chle ki hamri ek value false ha to hume dusri value nikalne ki zarurat hi ni ha.

```
int a = 5;
int b = 10;
int c = 15;

boolean e = (a > b) || (a < c);
```

Yha bhi samee ham jante ha ki || ke lia sirf ek hi value true chaiye, or agar kisi tarah pta chle ki meri ek value true ha to dusri value automatically true ho jaygi

Assignment Operators

```
// "=" assignment operator ha or "==" relational operator ha.

int a = 10;
int b = 10;
int c = 10;

// in assignment operator we can do, but we cant do with relational operators
a = b = c = 10;
System.out.println(a + b + c - a);
```

Ternary Operators

Ye operators three operands ke bich me lagta ha.

Operator precedence

Isko hum operators ka BODMAS bolege, ya batata ha ki agar bhot operators use ho re ha t phle kon operate krega or last me kon operate krega.

Highest						
++ (postfix)	-- (postfix)					
++ (prefix)	-- (prefix)	~	!	+ (unary)	- (unary)	(type-cast)
*	/	%				
+	-					
>>	>>>	<<				
>	>=	<	<=	instanceof		
==	!=					
&						
^						
&&						
?:						
->						
=	op=					
Lowest						